Mechanics of writing an economics research paper

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Spring 2026

1. General principles
2. Specifics
3. Next few weeks

1 General principles

2 Specifics

3 Next few weeks

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- Clearly state what assumptions are needed for your analysis to be valid
- Our evaluation of your paper will be based on how clearly you motivate and describe your work (not based on what you find)
  - Very importantly, it is completely ok to find "no effect"
  - Evaluation is based on clarity of presentation, interpretation
  - Evaluation *not* at all based on finding statistically significant estimates

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  - Body of paper: Tell readers what you are doing and finding
  - Conclusions: Recap what you did and what you found

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  - Conclusions: Recap what you did and what you found
- Oversell your results
  - Always caveat your results appropriately
  - *E.g.* "These results suggest that an x% increase in height is associated with a y% increase in wages"
  - Help readers understand both strengths and limitations of your analysis

Ways to try to achieve those goals

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   - Start early: this is a goal of the intermediate drafts
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2. Find (we can help) a paper to use as a model of how to structure your paper (doesn't have to be on the same topic)
3. Find an intended reader. E.g. ask your classmates or friends to read and comment on your paper

Ways to try to achieve those goals (continued)

Read and take (at least some of the) advice from others

- **Chaubey**: *The Little Book of Research Writing* (on syllabus)
- Shapiro: `https://shapiro.scholars.harvard.edu/sites/g/files/omnuum7731/files/2026-02/foursteps.pdf`
- McCloskey book (on syllabus)
- Cochrane: "Writing Tips for PhD Students"

Suggested elements and structure

- Short (one paragraph) abstract, around 100 words
- Introduction
- Possible intervening sections (vary across papers)
  - Theoretical framework (if relevant)
  - Background/institutional detail
  - *etc.*
- Data
- Empirical methods
- Estimation
- Conclusions

Today: More detail on each of these

1 General principles

2 Specifics

3 Next few weeks

Abstract

Briefly ($\sim$ 100 words):
Your question, what you do, what you find, what it means. (This from Williams (JPE, 2012)...)

Abstract

Do intellectual property (IP) rights on existing technologies hinder subsequent innovation? Using newly-collected data on the sequencing of the human genome by the public Human Genome Project and the private firm Celera, this paper estimates the impact of Celera's gene-level IP on subsequent scientific research and product development. Genes initially sequenced by Celera were held with IP for up to two years, but moved into the public domain once re-sequenced by the public effort. Across a range of empirical specifications, I find evidence that Celera's IP led to reductions in subsequent scientific research and product development on the order of 20 to 30 percent. Taken together, these results suggest that Celera's short-term IP had persistent negative effects on subsequent innovation relative to a counterfactual of Celera genes having always been in the public domain.

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  - Providing evidence on a relationship that is theoretically ambiguous?

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- Provide a (brief) roadmap for the rest of your paper

Introduction: First paragraph

Ideally: Big picture motivation to what you are doing in one paragraph

Innovation is a key driver of economic growth, but competitive markets may under-incentivize innovation due to the non-rival nature of ideas (Nelson, 1959; Arrow, 1962). Intellectual property (IP) rights, such as patents and copyrights, aim to incentivize innovation by allowing firms to capture a higher share of the social returns to their research investments. Traditional evaluations of the effectiveness of IP have focused on whether the prospect of obtaining IP rights stimulates the development of new technologies. However, in many markets technological change is cumulative, in the sense that product development results from several steps of invention and research. In markets where innovation is cumulative, the overall effectiveness of IP in promoting innovation also depends on a second, less studied question: do IP rights on *existing* technologies hinder *subsequent* innovation? The contribution of this paper is to provide empirical evidence on this second question by investigating how one form of IP on the human genome influenced subsequent scientific research and product development.

Introduction: Thought experiment

Ideally: Explain clearly the thought experiment behind your empirics

To fix ideas, suppose the firm Celera holds IP on a human gene, and that Pfizer discovers a genetic diagnostic test based on Celera's gene. Will Celera's IP discourage Pfizer from developing this test? In a perfect contracting environment with no transaction costs, Celera and Pfizer would negotiate a licensing agreement such that cumulative research is not hindered. However, transaction costs may cause licensing negotiations to break down, deterring some socially desirable research. While the theoretical literature on this question is well-developed (Scotchmer (1991), Green and Scotchmer (1995), Bessen (2004)), empirical evidence is scarce.

Introduction: Possible outline

- Your question
- Why we care
- What you do
- What you find
- (Brief) How this relates to what others have found
- (Brief) What this means
- Roadmap for paper (see below)

The paper proceeds as follows. Section 1 provides a brief scientific background and a description of my data construction. Section 2 describes the empirical context. Section 3 presents the empirical results. Section 4 concludes by linking my empirical results back to previous theoretical and empirical analyses in an attempt to interpret which types of transaction costs were likely most important in this context.

Background/institutional detail (if relevant)

- Key details needed to understand your analysis
  - Inform the key details of your empirical strategy
  - Discuss background relevant for your assumptions
- *Not* a general history of the legislation (for example)

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  - Described sufficiently clearly and completely that the reader could replicate your estimates
  - Text of your paper should contain the key details
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- Present summary statistics in a table (means, standard deviations)

Summary Statistics for Gene-Level Data

	mean	med	std dev	min	max
Panel A: Sequencing & Celera's IP					
year sequence disclosed	2003.0	2001	3.551	1999	2009
1(Celera gene)	0.060	0	0.238	0	1
Panel B: Outcome variables					
publications in 2001-2009	2.197	0	9.133	0	231
1(known, uncertain phenotype)	0.453	0	0.498	0	1
1(known, certain phenotype)	0.081	0	0.273	0	1
1(used in any diagnostic test)	0.060	0	0.238	0	1
$N = 27{,}882$					

Notes: See Table 2 in paper.

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- Discuss how you could test those assumptions (and do the test if possible!)

Empirical methods: Example

The concern with interpreting these simple differences in means is that they could reflect a negative effect of Celera's IP on subsequent research, or could instead reflect selection if Celera's genes had lower inherent potential for follow-on research. In order to address this concern, Table 3 formalizes the basic comparison from Table 1 in a regression framework that allows me to investigate the robustness of these patterns. For gene g, I estimate the following:

$$(outcome)_g = \beta(celera)_g + \lambda'(covariates)_g + \varepsilon_g.$$

The coefficient on the "*celera*" variable is the main estimate of interest. I show estimates from ordinary-least-squares (OLS) models, and report heteroskedasticity-robust standard errors.

Estimation

- Present your results in clear, well-labeled tables and figures
  - Goal: self-contained tables/figures (clear titles and extensive notes)
  - Look at a journal article to see how tables are laid out
  - Include sample sizes in each column (watch for sample size migration!)
  - Show standard errors, not t-statistics
  - Stars? Your call.

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- Describe *and interpret* your results in the text
  - Both economic significance (magnitude) and statistical significance (or lack thereof)

Estimation: Example

Table 3: Cross-Section Estimates: Impact of Celera's IP on Innovation Outcomes

	(1)	(2)	(3)
Panel A: publications in 2001-2009			
mean = 2.197			
celera	-0.877	-0.328	-0.264
	(0.177)***	(0.099)***	(0.107)***

Estimation: Example

indicator variables for year of disclosure	yes	yes	yes
number of publications in each year 1970-2000	no	yes	yes
detailed cytogenetic & molecular covariates	no	no	yes
N	27,882	27,882	16,485

Notes: Gene-level observations. All estimates are from ordinary-least-squares (OLS) models. Samples in Columns (1) and (2) include all genes; sample in Column (3) includes all genes with non-missing cytogenetic and molecular covariates. Robust standard errors shown in parentheses. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$. *"Celera"*: 0/1, =1 for a Celera gene. *Indicator variables for year of disclosure*: 0/1 indicator variables for the year sequence for the gene was disclosed. *Number of publications in each year 1970-2000*: count variables for the number of publications on each gene in each year from 1970 to 2000. *Detailed cytogenetic & molecular covariates*: 0/1 indicator variables for the chromosome (1-22, X, or Y) and arm (p or q) on which a gene is located; continuous variables for region, band, subband, start base pair, and end base pair; and 0/1 indicator variables for the orientation of the gene on the genome assembly (plus or minus). See text and the online appendix for more detailed data and variable descriptions.

Estimation: Interpreting magnitudes

For brevity, I focus on interpreting the magnitudes of the point estimates in Column (1). The estimate in Panel A implies Celera genes had about 0.88 fewer publications from 2001 to 2009, a decline on the order of 40% of the mean number of publications for genes over that time period.[19] The estimate in Panel B implies a 16 percentage point reduction in the probability of having a known, uncertain phenotype link, a decline on the order of 35% relative to the sample mean. The estimate in Panel C implies a 2.7 percentage point reduction in the probability of having a known, certain phenotype link, a decline on the order of 33% of the sample mean. The estimate in Panel D implies a 2.3 percentage point reduction in the probability of a gene being used in any currently available diagnostic test, a decline on the order of 38% of the sample mean.

Conclusions

- Tie elements together
- Summarize results (short but sweet)
- Discuss main limitations
- Discuss policy relevance
- Mention directions for future research

Conclusions

Can and should be short (really). (But one possibility for extending the below would be to acknowledge areas where further research, building on the current study and its limitations, would be valuable.)

4.4 Concluding remarks

Intellectual property (IP) is a widely-used policy lever for promoting innovation, yet relatively little is known about how IP on existing technologies affects subsequent innovation. The sequencing of the human genome provides a useful empirical context, generating variation in IP across a relatively large group of *ex ante* similar technologies. Across a range of empirical specifications, I find evidence that Celera's IP led to reductions in subsequent scientific research and product development on the order of 20 to 30 percent.

The overall welfare effects of IP depend on other factors, including the provision of dynamic incentives for developing new technologies.[26] From a policy perspective, these results suggest that, holding Celera's entry constant, an alternative lump-sum reward mechanism may have had social benefits relative to Celera's chosen form of IP.

Your .do and .log file (or R equivalent)

- You will need to turn these in
- Please comment your code to make it easier for us to follow
- Please print these and append them into one PDF file for submission

1 General principles

2 Specifics

3 Next few weeks

Remainder of semester

- 4/13: A7: revised draft of data/methods with summary statistics, outline, and results due, along with document detailing changes to A5 and A6
- 4/27: A8: first draft of paper due
- 4/29 and 5/14: A9 final presentations
- 5/11: A10: final draft of paper (with revisions to comments from A8) due